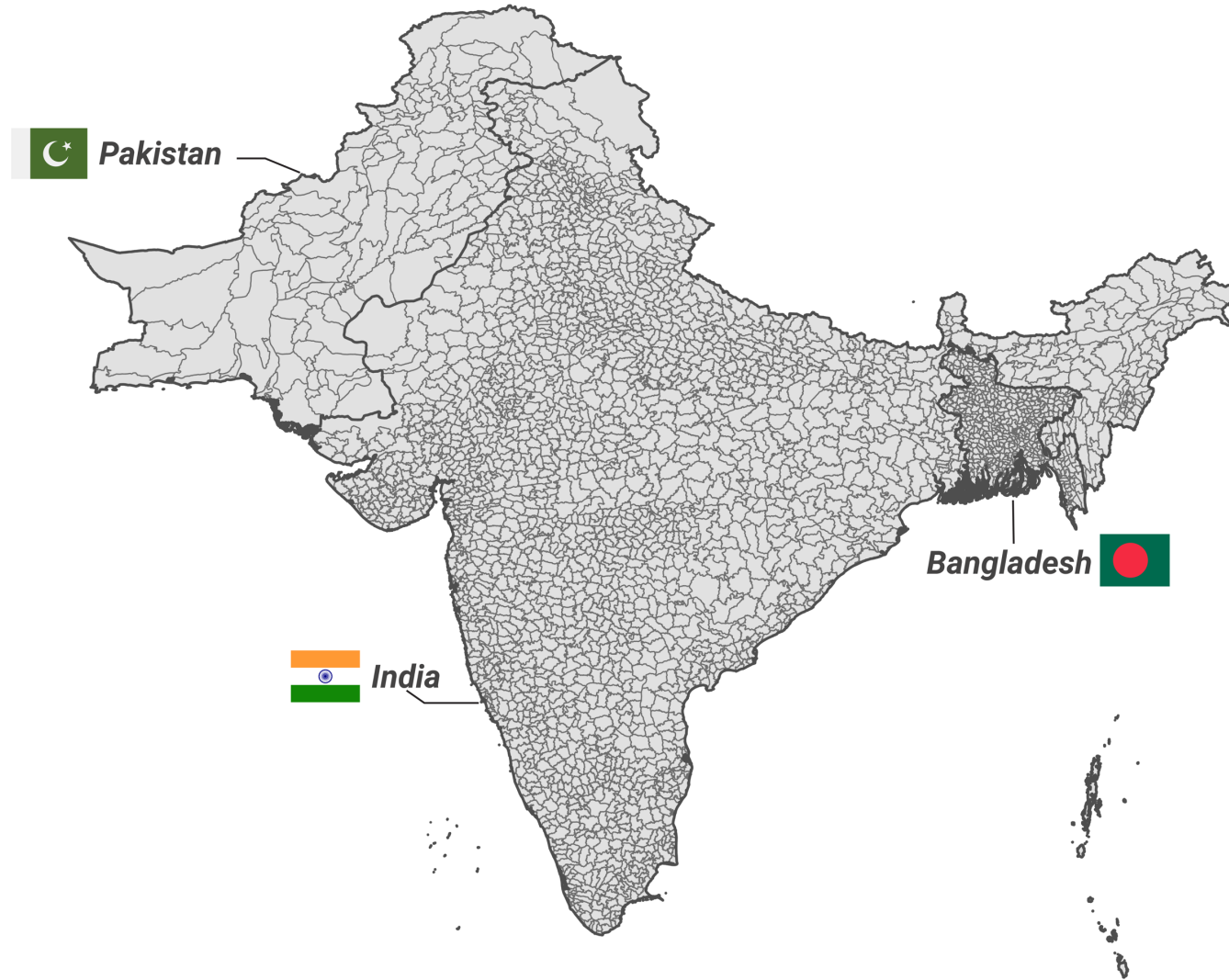


A white humanoid robot is seated at a desk, typing on a laptop. The robot has a sleek, modern design with visible joints and a friendly-looking face. The laptop screen displays a code editor with syntax-highlighted text. The background is a blurred office environment with large windows.

Programming for AI

Usman Nazir

Review



Review

Virtual-Try-On

like

3.02k

Running on CPU UPGRADE

App

Kolors Virtual Try-On in the Wild

Tech Report

Kolors

Official Website

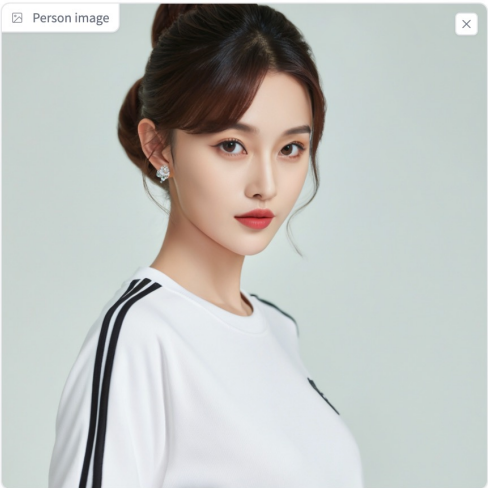
CN

Official Website

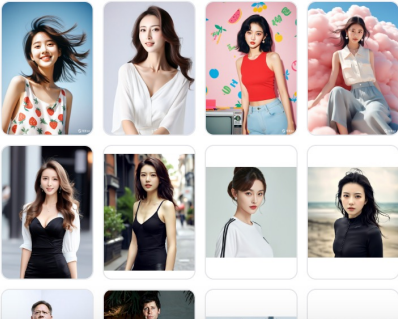
EN

Disclaimer: The Kolors-Virtual-Try-On Demo on HuggingFace space is free for trials. Any solicitation for payment based on the free features we provide on HuggingFace Space is a fraudulent act. Beware of scams and do not fall victim to deceit.

Step 1. Upload a person image



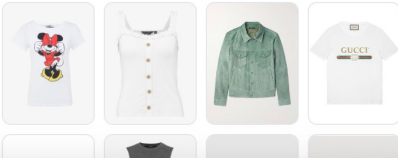
Examples



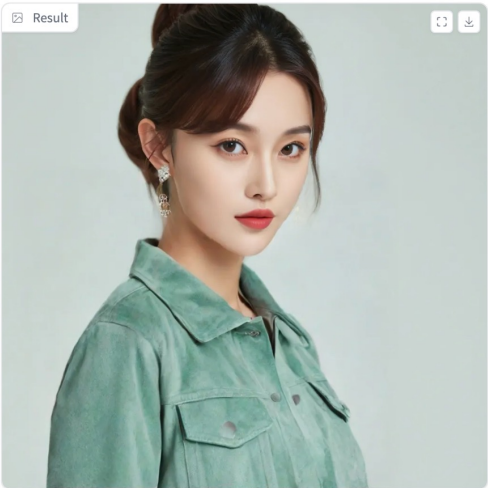
Step 2. Upload a garment image



Examples



Step 3. Press "Run" to get try-on results



Seed

0

Random seed

Seed used

35353

Response

Success

Run

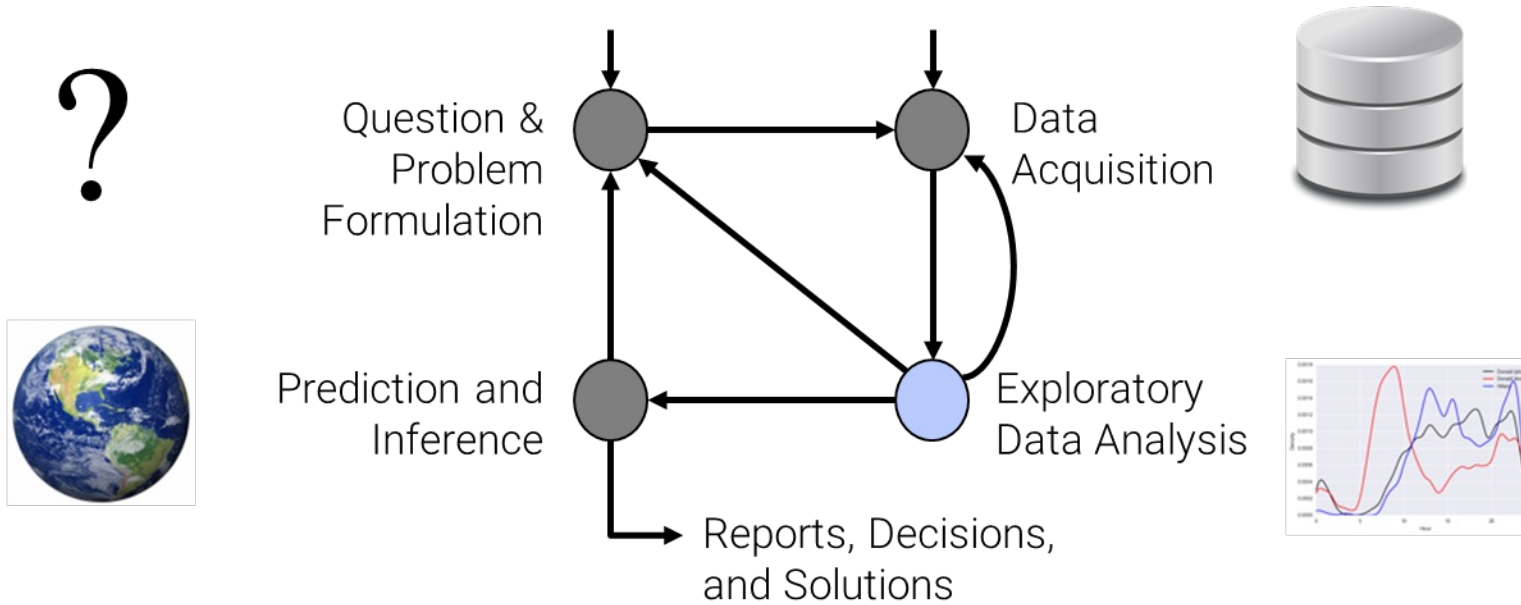
Review

Machine Learning is **function mapping between input (A) and output (B).**

Size of house (Square feet)	Price (Lakhs)
523	115
645	150
708	210
1034	280
2290	355
2545	440

Agenda

- What is Python?



Python Definition

Python is an interpreted general purpose (or multi-purpose) high level programming language with easy syntax and dynamic semantics.

Python Definition

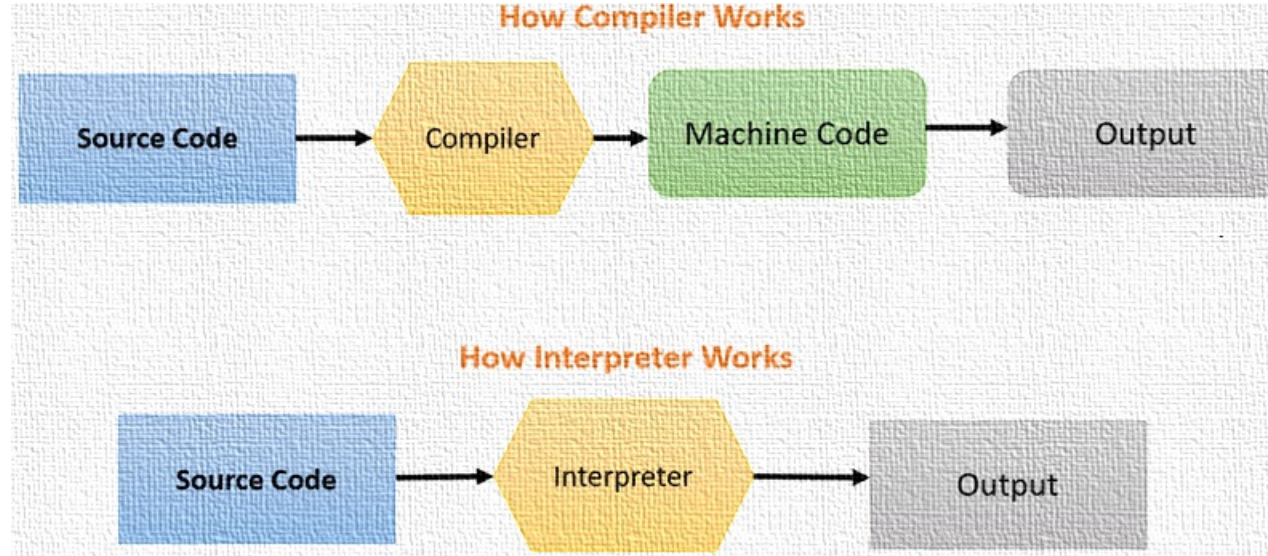
Python is an **interpreted** **general purpose** (or multi-purpose) **high level programming** language with **easy syntax** and **dynamic semantics**.

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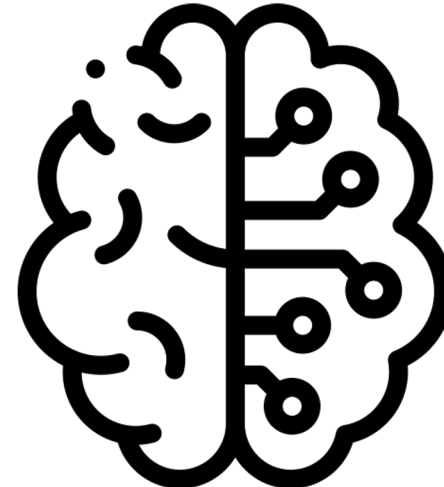
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Isn't specialized for specific domain

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level programming language with **easy syntax** and **dynamic**
semantics.

Python Definition

Python is an **interpreted** **general purpose** (or multi-purpose) **high level programming** language with **easy syntax** and **dynamic semantics**.

```
100100
10000000011
101010101010
0000000001111
```

Low level language

VS

```
if(i<5)
{
printf("I am true block ");
}
else{
printf("I am false block");
}
```

High level language

Python Definition

Python is an **interpreted** **general purpose** (or multi-purpose) **high level programming** language with **easy syntax** and **dynamic semantics**.

Python Definition

Python is an **interpreted** general purpose (or multi-purpose) **high level programming** language with **easy syntax** and **dynamic semantics**.

```
int x = 10;  
int y = 5;  
printf("%d", x+y);
```

C code

Python Definition

Python is an **interpreted** general purpose (or multi-purpose) **high level programming** language with **easy syntax** and **dynamic semantics**.

```
int x = 10;  
int y = 5;  
printf("%d", x+y);
```

C code

```
x = 10  
y = 5  
print(x+y)
```

Python code

Python Definition

Python is an **interpreted** **general purpose** (or multi-purpose) **high level programming** language with **easy syntax** and **dynamic semantics**.

Python Definition

Python is an **interpreted** general purpose (or multi-purpose) **high level programming** language with **easy syntax** and **dynamic semantics**.

```
int x = 10;  
int y = 5;  
printf("%d", x+y);  
x = 10.78;
```

C code

Statically typed

```
x = 10  
y = 5  
print(x+y)  
x = 10.78
```

Python code

Dynamically typed

Summary

Python is an **interpreted** **general purpose** (or multi-purpose) **high level programming** language with **easy syntax** and **dynamic semantics**.

Introduction to Data Science

Python Refresher



Usman Nazir

Agenda

- What is Python?
- Problems
 - Make a simple calculator

Problem 0: Make a simple calculator

- Hint

- Take input from user e.g. `first = input("Enter first number: ")`

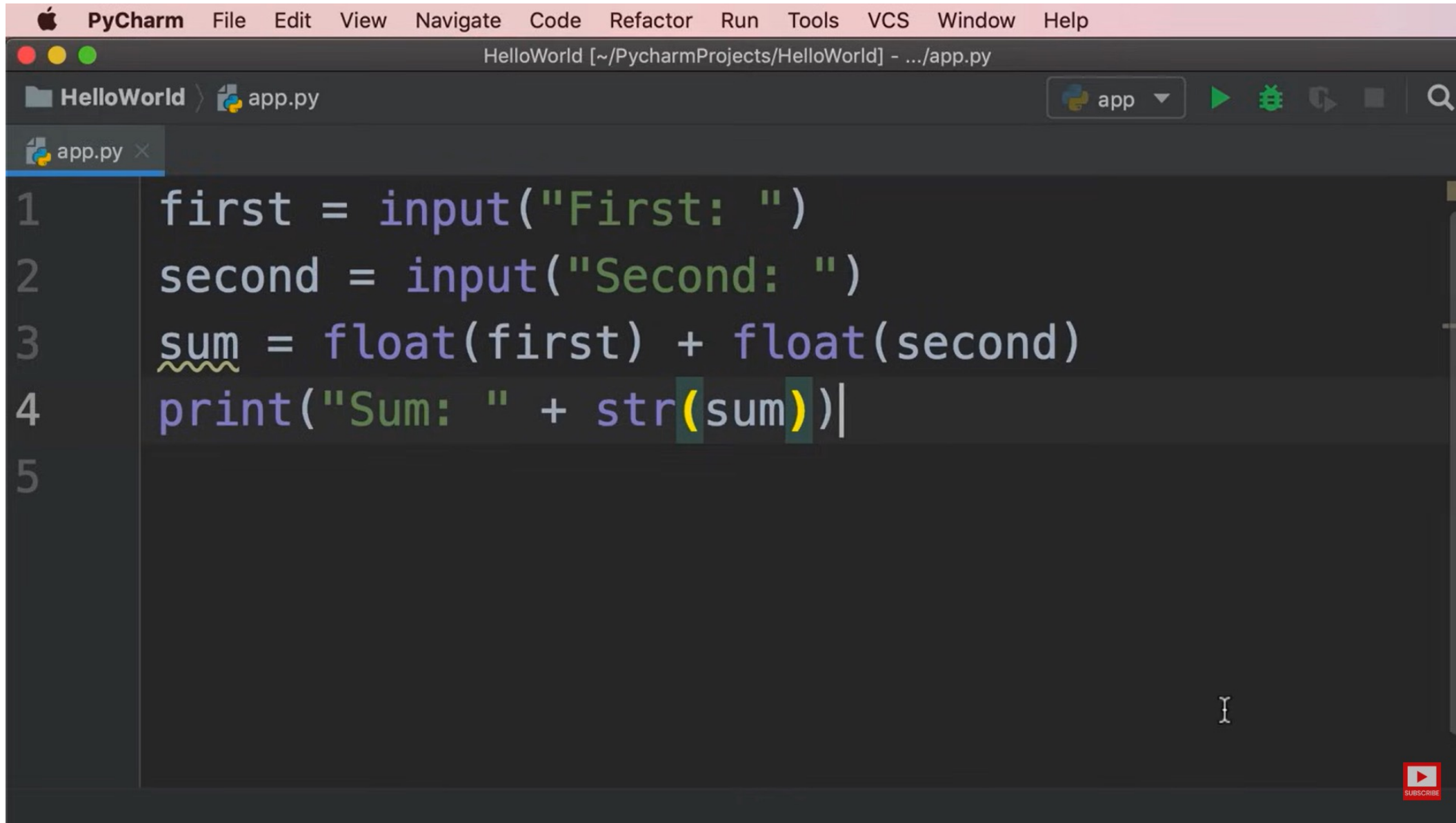
```
First: 10.1
```

```
Second: 20
```

```
Sum: 30.1
```

```
I
```


Problem 0: Make a simple calculator



The image shows a PyCharm IDE window with a menu bar (File, Edit, View, Navigate, Code, Refactor, Run, Tools, VCS, Window, Help) and a title bar (HelloWorld [~/PycharmProjects/HelloWorld] - .../app.py). The file explorer on the left shows the 'HelloWorld' directory containing 'app.py'. The main editor area displays the following Python code in 'app.py':

```
1 first = input("First: ")
2 second = input("Second: ")
3 sum = float(first) + float(second)
4 print("Sum: " + str(sum))
5
```

The code is syntactically highlighted, with the variable `sum` underlined on line 3. A cursor is visible at the end of line 4. In the bottom right corner, there is a small red YouTube 'SUBSCRIBE' button.

Summary

- What is Python?
- Problems
 - Make a simple calculator

Introduction to Data Science

Python Refresher

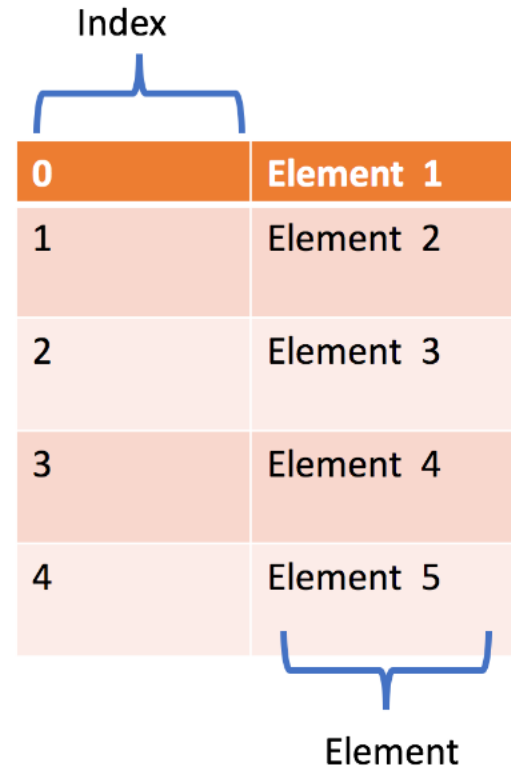


Usman Nazir

Agenda

- What is Python?
- Problems
 - Make a simple calculator
 - Area of Box

Lists



A diagram illustrating a list structure. It consists of a table with two columns. The first column is labeled 'Index' with a blue bracket above it. The second column is labeled 'Element' with a blue bracket below it. The table contains five rows of data. The first row has an orange header with '0' and 'Element 1'. The subsequent rows have a light pink background and contain '1' and 'Element 2', '2' and 'Element 3', '3' and 'Element 4', and '4' and 'Element 5' respectively.

Index	
0	Element 1
1	Element 2
2	Element 3
3	Element 4
4	Element 5
Element	

[Element 1 , Element 2 , Element 3 , Element 4, Element 5]

Index 0 1 2 3 4

Lists

List Operations

We can also perform slicing in lists. For example, if we want the last two elements, we use the following command:

```
[1] # Sample List  
  
L = ["Michael Jackson", 10.1, 1982, "MJ", 1]  
L  
  
['Michael Jackson', 10.1, 1982, 'MJ', 1]
```

L = ["Michael Jackson", 10.1, 1982, "MJ", 1]

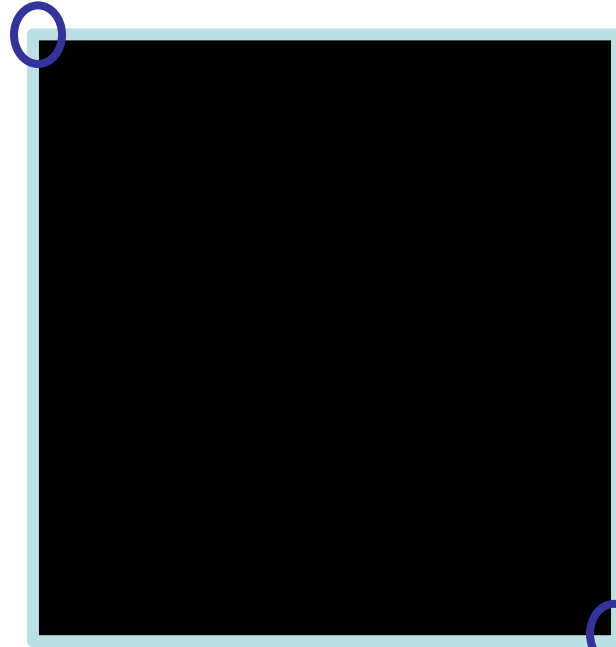
0	1	2	3	4
---	---	---	---	---

L[0] = ?

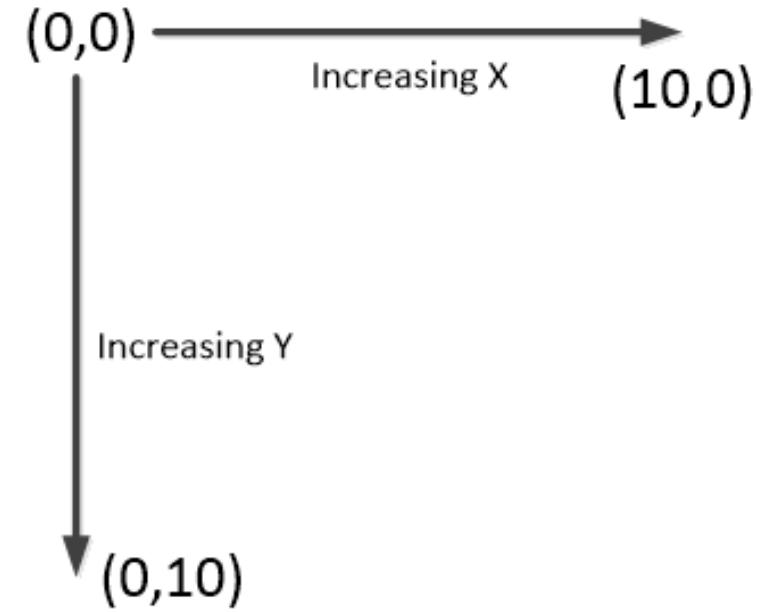
L[-1] = ?

Problem 1: Area of Box

$(x1, y1)$



$(x2, y2)$



Summary

- What is Python?
- Problems
 - Make a simple calculator
 - Area of Box

Introduction to Data Science

Python Refresher

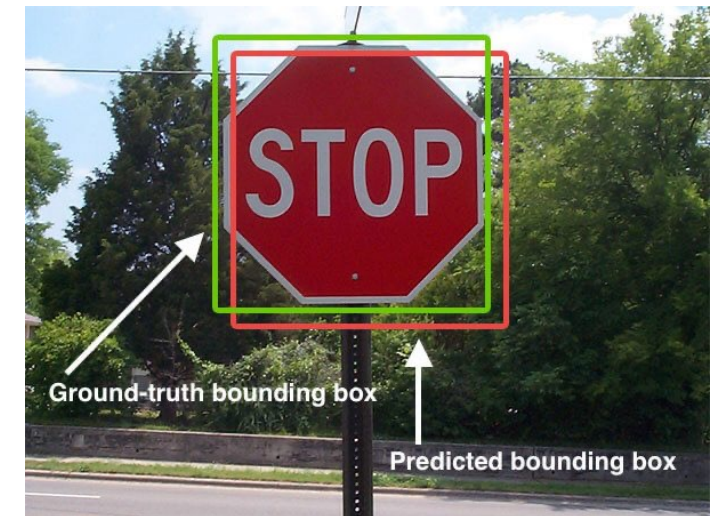
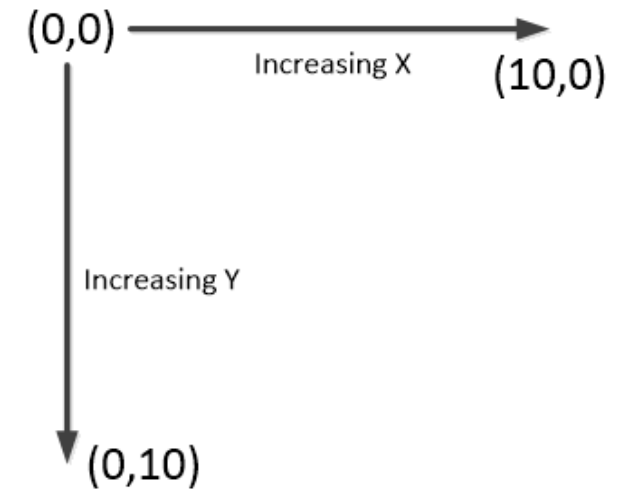
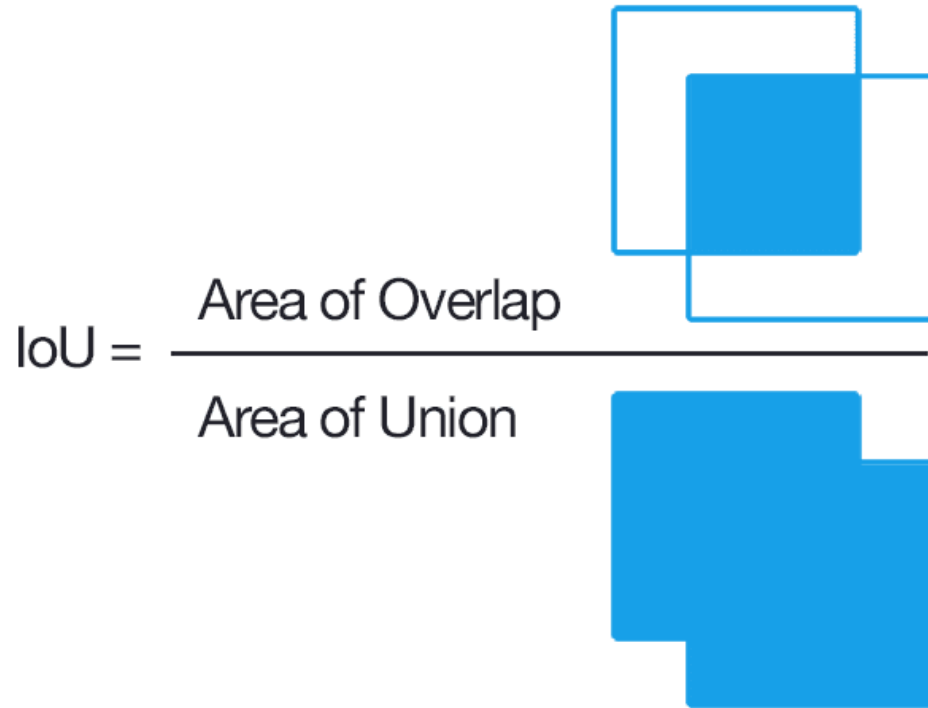


Usman Nazir

Agenda

- What is Python?
- Problems
 - Make a simple calculator
 - Area of Box
 - Intersection over Union (IoU)

Problem 2: Intersection over Union



Summary

- What is Python?
- Problems
 - Make a simple calculator
 - Area of Box
 - Intersection

Introduction to Data Science

Python Refresher

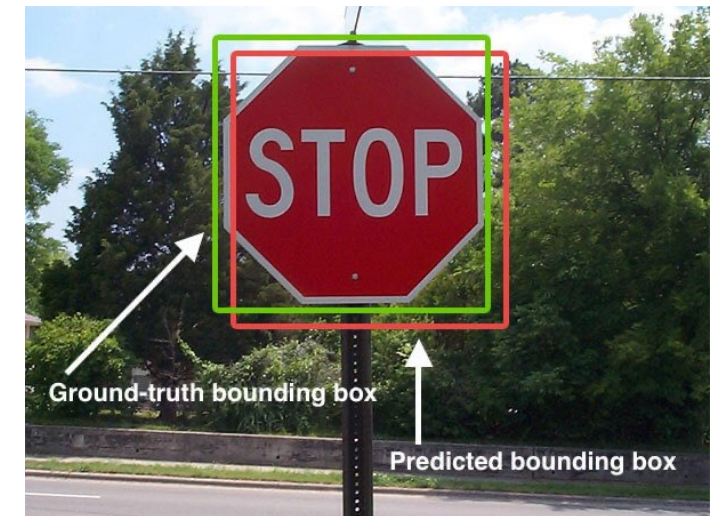
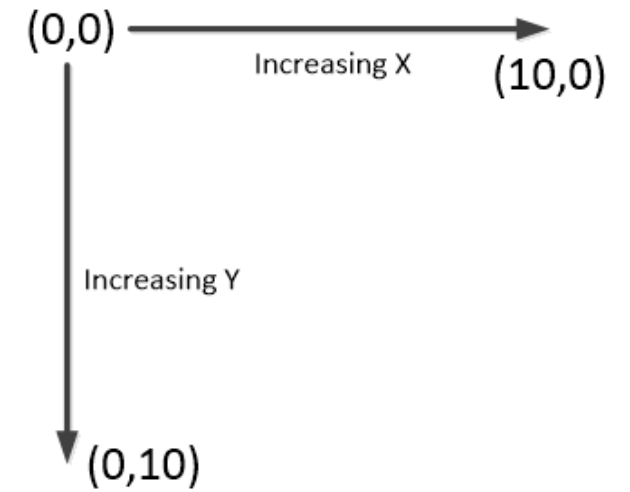
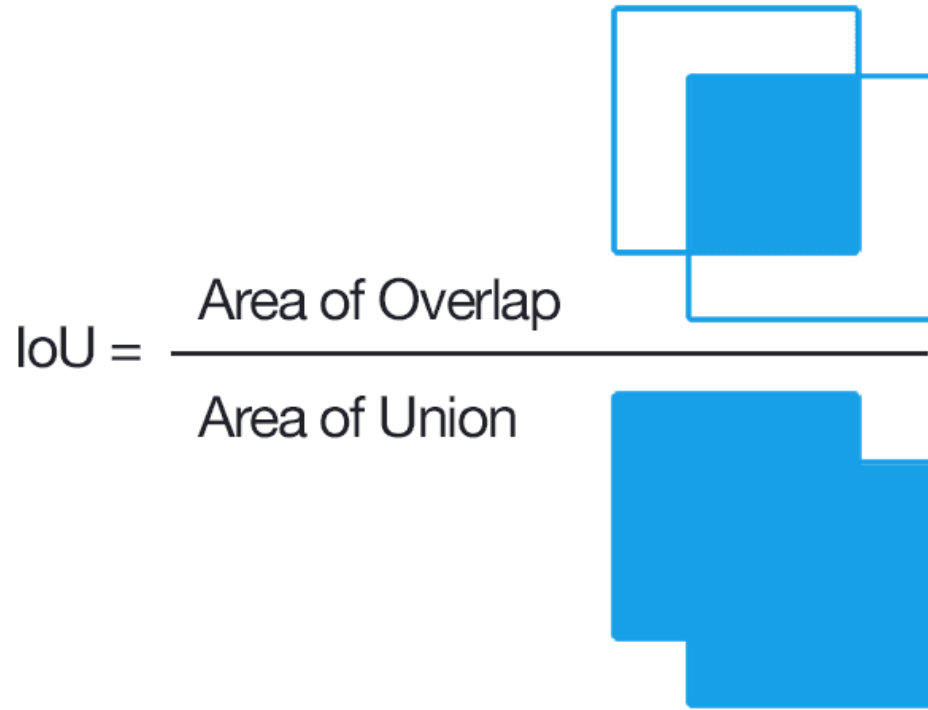


Usman Nazir

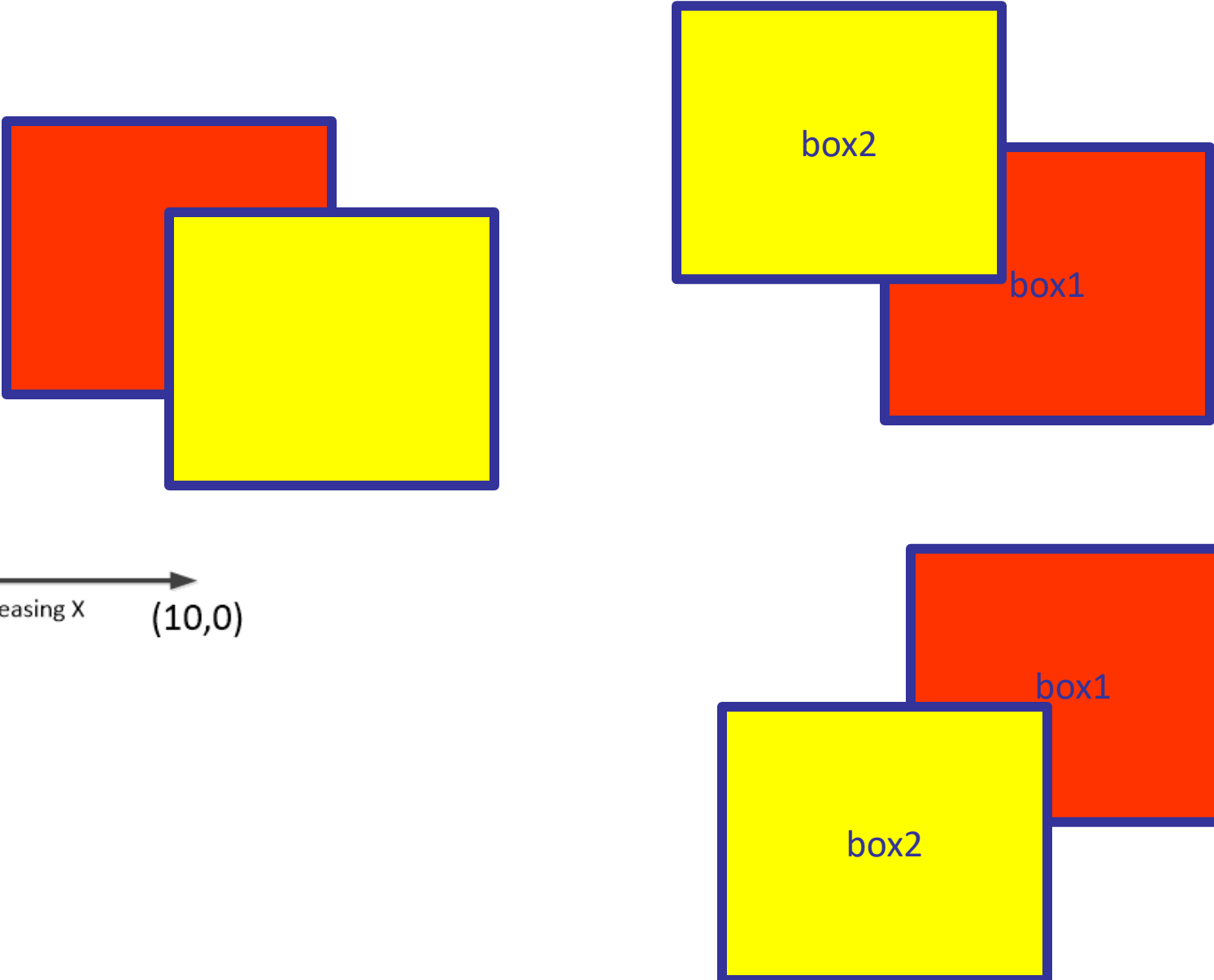
Agenda

- What is Python?
- Problems
 - Make a simple calculator
 - Area of Box
 - Intersection over Union (IoU)

Problem 2: Intersection over Union



Intersection Area



Summary

- What is Python?
- Problems
 - Make a simple calculator
 - Area of Box
 - Intersection
 - Union
 - IoU